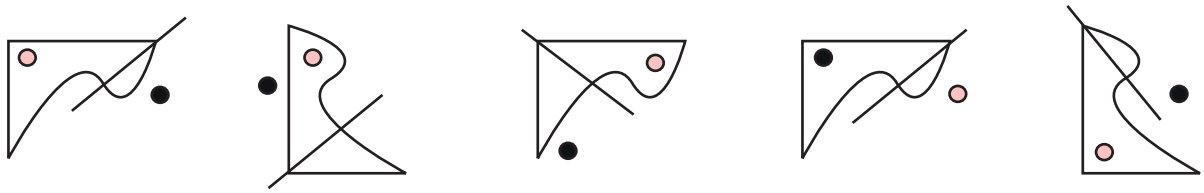
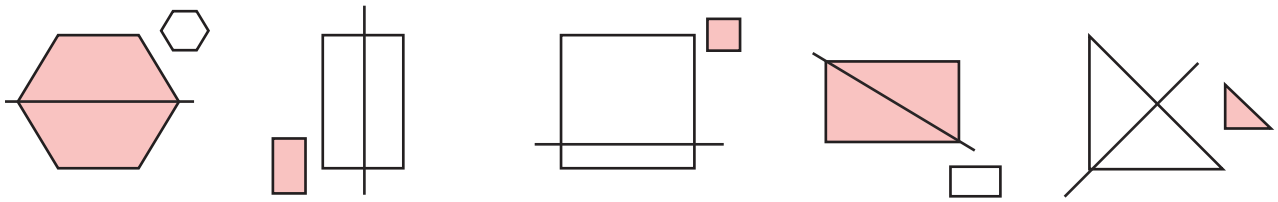
Which figure is the odd one out? Tick the correct box.

1



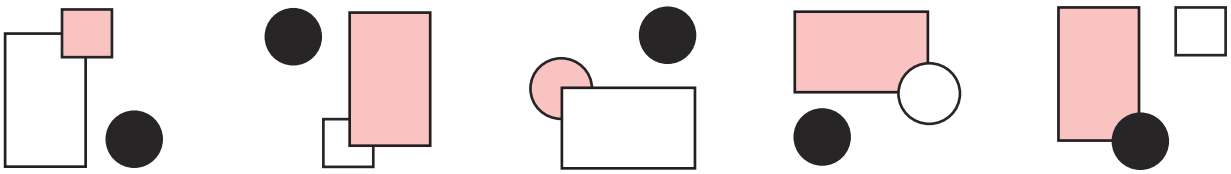
a ☐ b ☐ c ☐ d ☐ e ☐

2



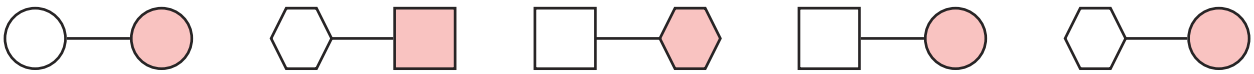
a ☐ b ☐ c ☐ d ☐ e ☐

3



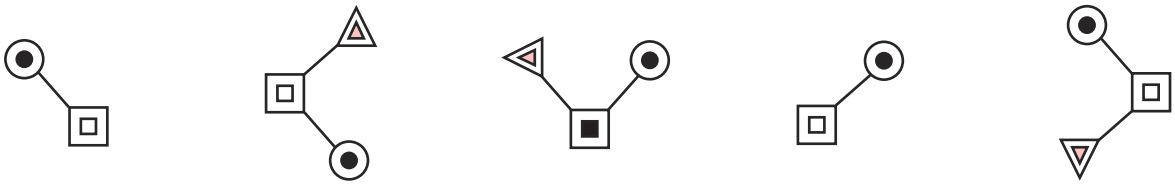
a ☐ b ☐ c ☐ d ☐ e ☐

4



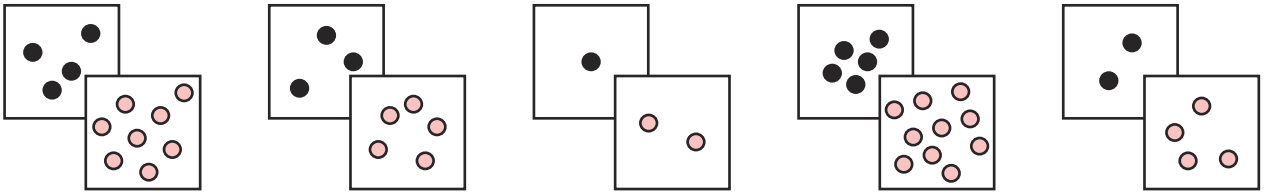
a ☐ b ☐ c ☐ d ☐ e ☐

5



a ☐ b ☐ c ☐ d ☐ e ☐

6

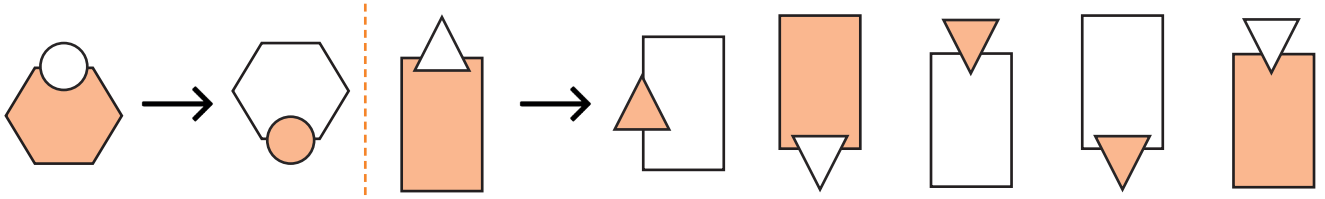


a ☐ b ☐ c ☐ d ☐ e ☐

Look at how the first figure changes to become the second figure.

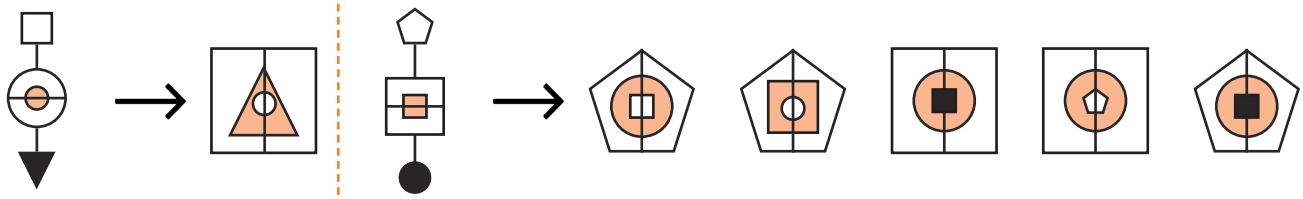
Tick the figure that the third figure will become if it is changed in the same way.

1



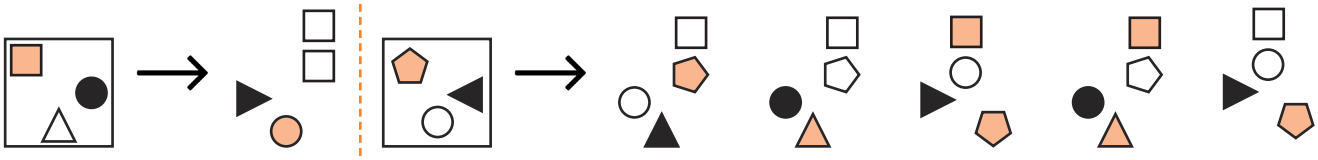
a ☐ b ☐ c ☐ d ☐ e ☐

2



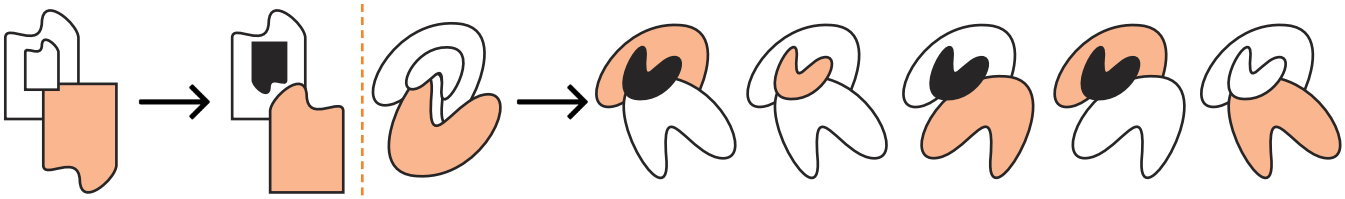
a ☐ b ☐ c ☐ d ☐ e ☐

3



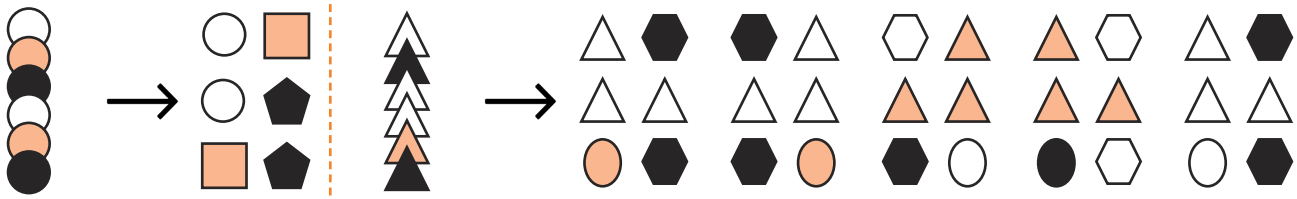
a ☐ b ☐ c ☐ d ☐ e ☐

4



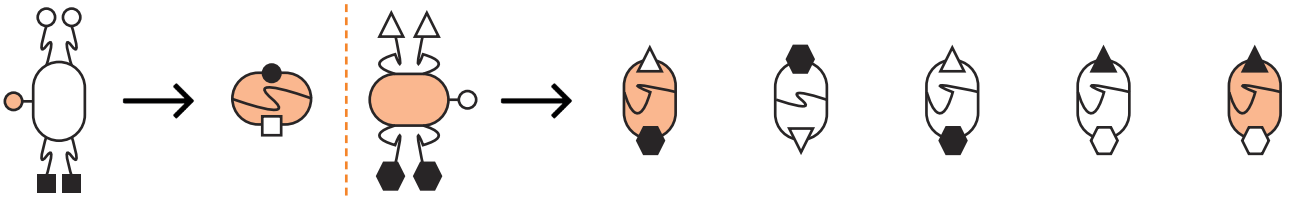
a ☐ b ☐ c ☐ d ☐ e ☐

5



a ☐ b ☐ c ☐ d ☐ e ☐

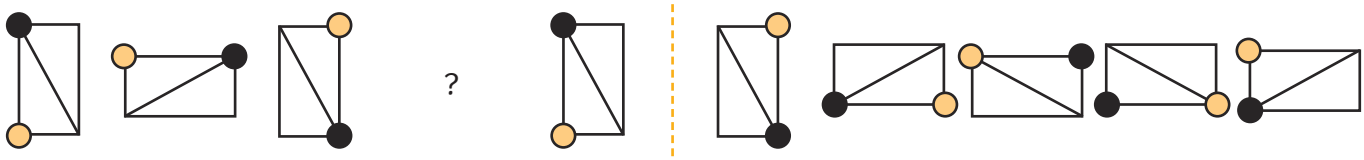
6



a ☐ b ☐ c ☐ d ☐ e ☐

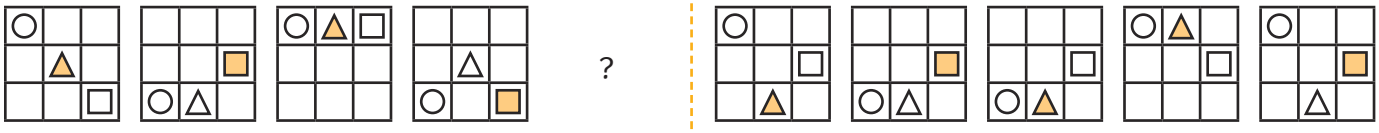
A figure is missing from each of these sequences. Tick the figure that is missing.

1



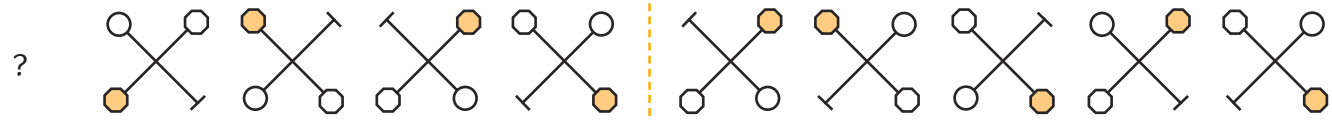
a ☐ b ☐ c ☐ d ☐ e ☐

2



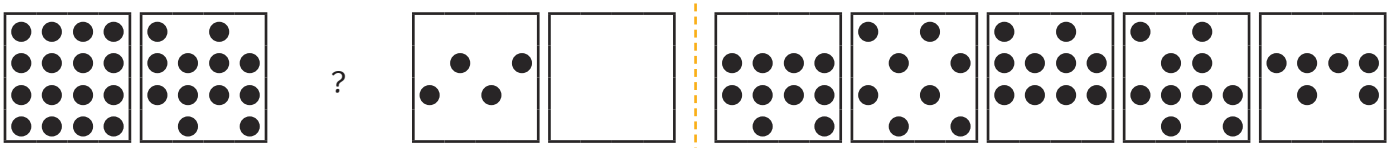
a ☐ b ☐ c ☐ d ☐ e ☐

3



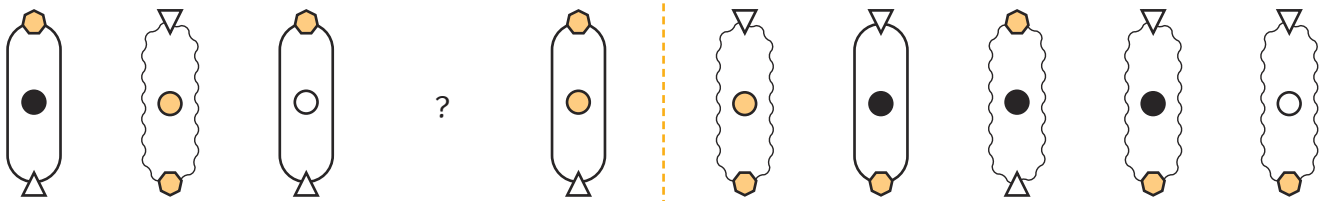
a ☐ b ☐ c ☐ d ☐ e ☐

4



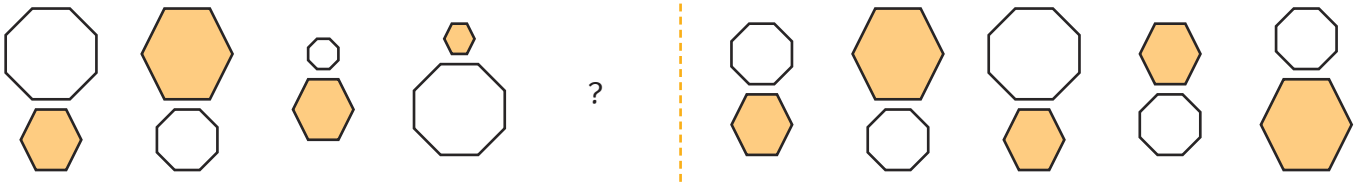
a ☐ b ☐ c ☐ d ☐ e ☐

5



a ☐ b ☐ c ☐ d ☐ e ☐

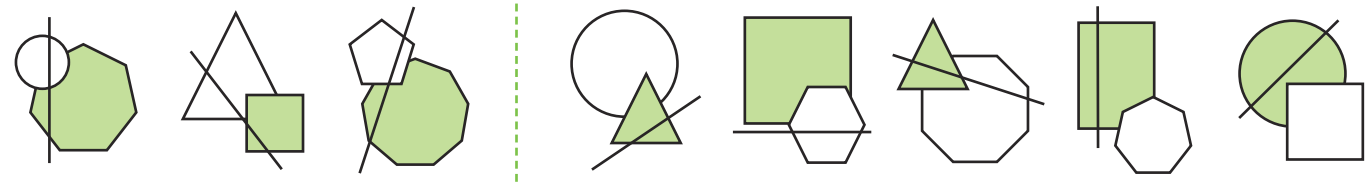
6



a ☐ b ☐ c ☐ d ☐ e ☐

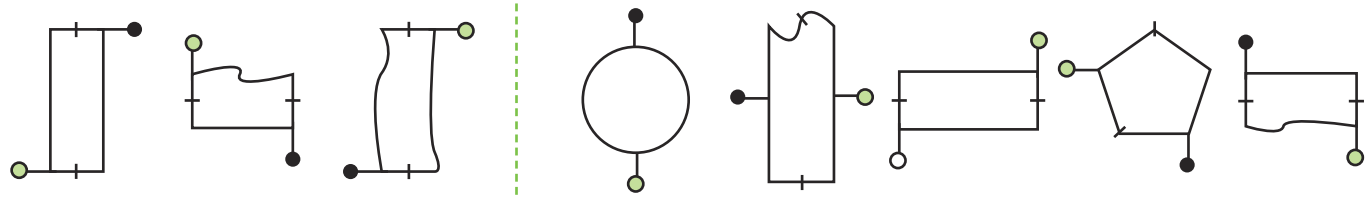
In each question there are **three** figures on the left which are alike in one or more ways.
Choose which of the **five** options on the right is most like the figures on the left.

1



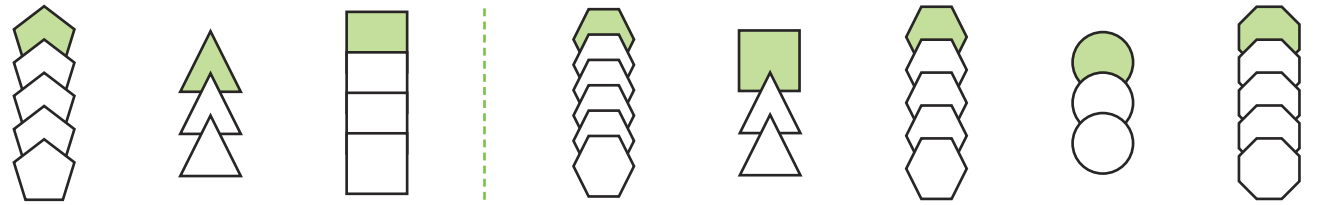
a ☐ b ☐ c ☐ d ☐ e ☐

2



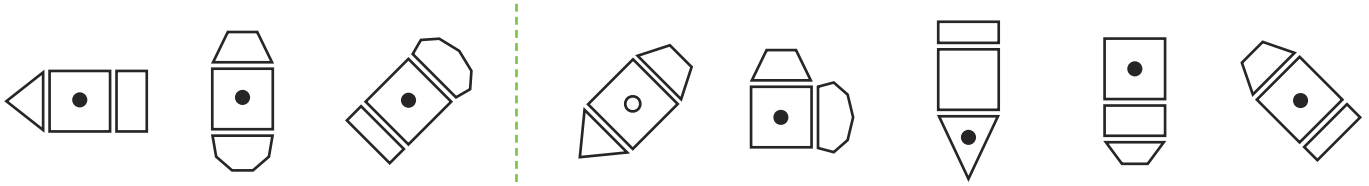
a ☐ b ☐ c ☐ d ☐ e ☐

3



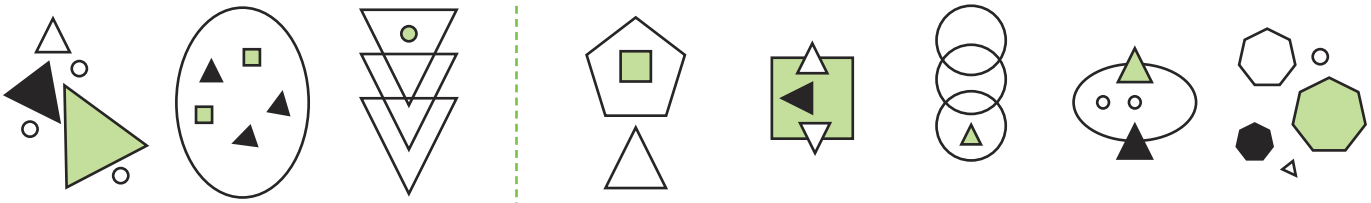
a ☐ b ☐ c ☐ d ☐ e ☐

4



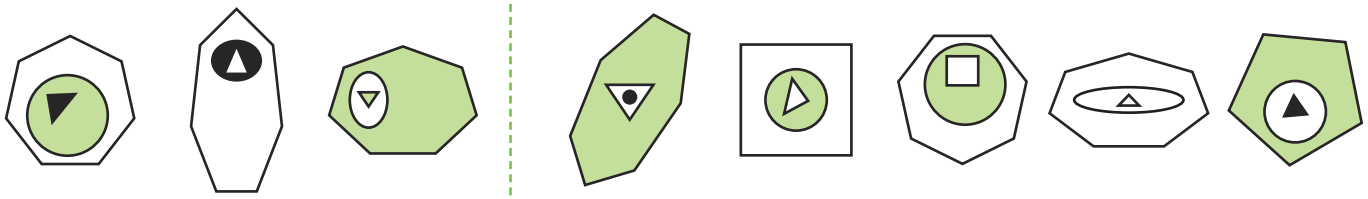
a ☐ b ☐ c ☐ d ☐ e ☐

5



a ☐ b ☐ c ☐ d ☐ e ☐

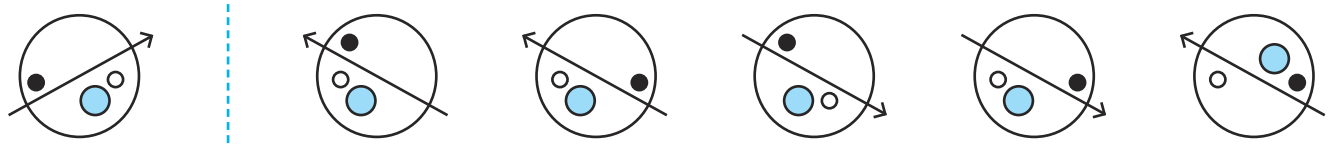
6



a ☐ b ☐ c ☐ d ☐ e ☐

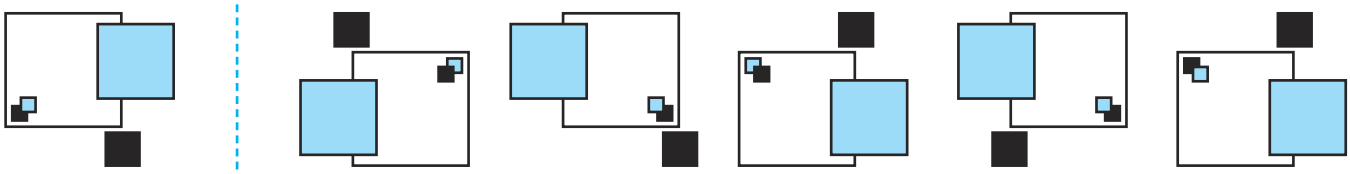
Which of the five options on the right is the reflection of the figure on the left?
Use the mirror line provided.

1



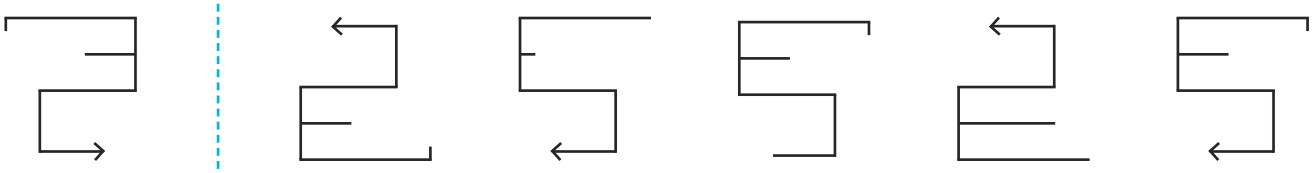
a ☐ b ☐ c ☐ d ☐ e ☐

2



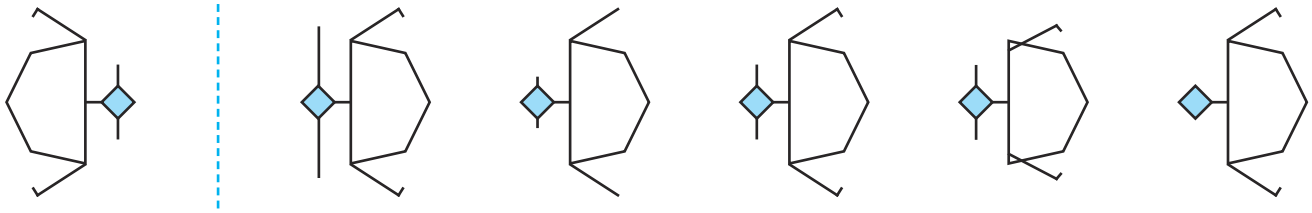
a ☐ b ☐ c ☐ d ☐ e ☐

3



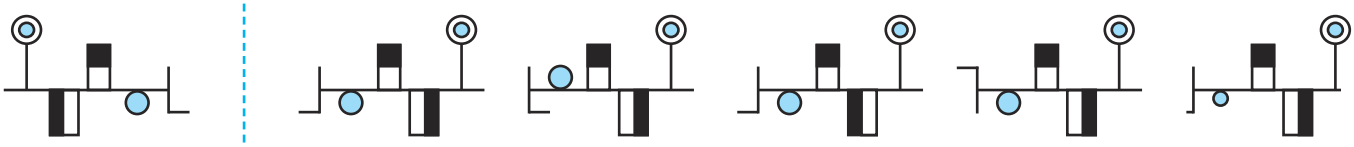
a ☐ b ☐ c ☐ d ☐ e ☐

4



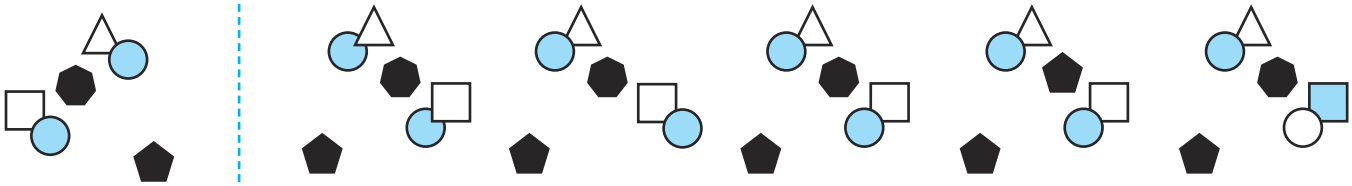
a ☐ b ☐ c ☐ d ☐ e ☐

5



a ☐ b ☐ c ☐ d ☐ e ☐

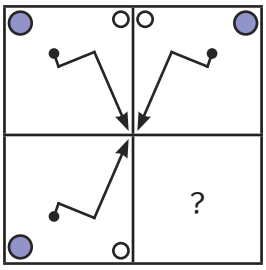
6

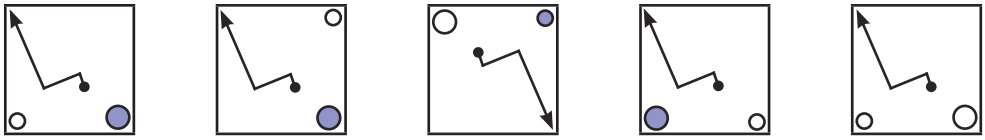


a ☐ b ☐ c ☐ d ☐ e ☐

Which square completes the grid? Tick the correct answer.

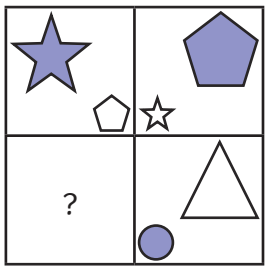
1

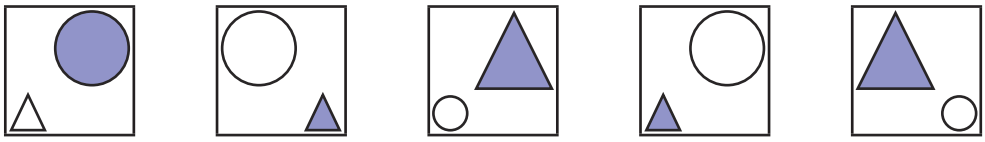




a ☐ b ☐ c ☐ d ☐ e ☐

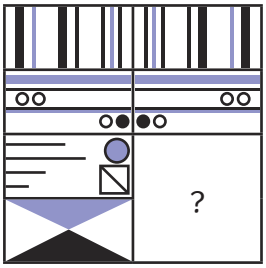
2





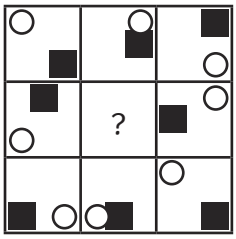
a ☐ b ☐ c ☐ d ☐ e ☐

3



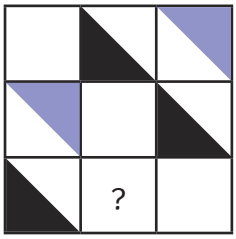
a ☐ b ☐ c ☐ d ☐ e ☐

4



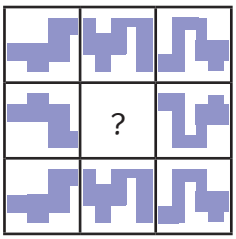
a ☐ b ☐ c ☐ d ☐ e ☐

5



a ☐ b ☐ c ☐ d ☐ e ☐

6



a ☐ b ☐ c ☐ d ☐ e ☐

In each question there are **five** shapes on the left. **Four** have a pair of code letters that describe them but the **last** shape is missing these. Work out what the code letters mean and then tick the code that is missing.

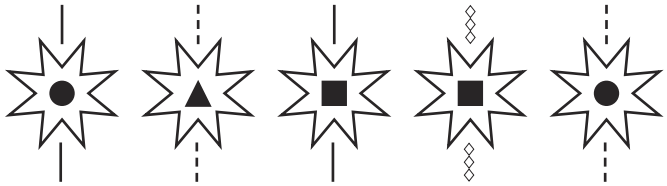
1

A R B S A S C T ? ?

C R B R A T C S B T

a ☐ b ☐ c ☐ d ☐ e ☐

2

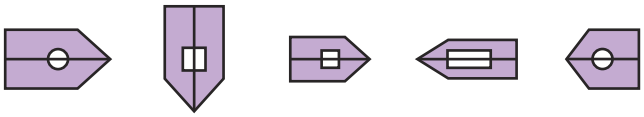


X E Y F Z E Z G ? ?

X F Y G Y E Z F X G

a ☐ b ☐ c ☐ d ☐ e ☐

3

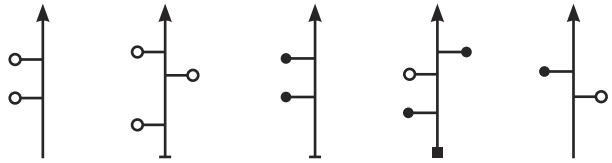


H Q I R H R J S ? ?

J R I Q J Q I S H R

a ☐ b ☐ c ☐ d ☐ e ☐

4

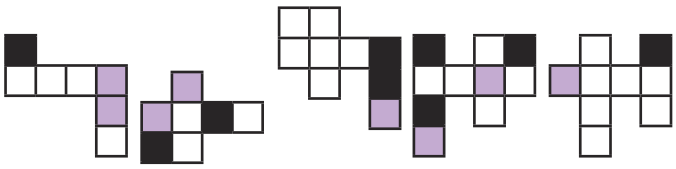


L T M T M U N V ? ?

L U M V N T N U L V

a ☐ b ☐ c ☐ d ☐ e ☐

5

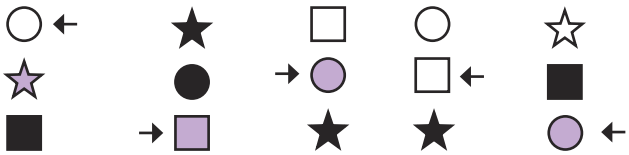


C S C R D R E T ? ?

D T E S E R D S C T

a ☐ b ☐ c ☐ d ☐ e ☐

6



B X C Y A Z B Z ? ?

A Y A X C Z B Y C X

a ☐ b ☐ c ☐ d ☐ e ☐

1 d

In all other figures, the pink circle is inside the shape and the black circle is outside the shape.

2 c

In all other figures, the line divides the larger shape into two equal parts.

3 e

In all other figures, the black circle does not touch any other shape, while the other small shape intersects with the larger shape.

4 a

In all other figures, the two shapes connected by the line are different shapes.

5 c

In all other figures, the smaller square inside the larger square is white.

6 b

In all other figures, the number of black circles are exactly half the number of pink circles.

1 d

- The figure has been turned upside down.
- The colours of the shapes have been swapped.

2 a

- The top shape in the first figure becomes the largest shape in the second figure.
- The bottom shape becomes the second largest shape and turns orange.
- The small shape in the centre of the middle shape stays the same size but turns white. The middle shape disappears.
- The line that divides the middle shape is turned 90° and now divides the largest shape.

3 b

- The three shapes in the box rotate clockwise by 90°.
- The colours of the shapes also rotate - each colour moves clockwise to the next shape.
    - The white shape turns black.
    - The orange shape turns white.
    - The black shape turns orange.
- The outside box shrinks and is placed above the top right shape.

4 c

- The orange shape is rotated 180°.
- The smaller white shape is rotated 180° and turns black.

5 a

- The white shapes stay the same shape, the orange shapes and the black shapes change into different shapes.
- The shapes are reordered in the following pattern/direction.

6 d

- The oval is turned 90°.
- The oval becomes the same colour as the circle that is attached to it with a straight line. This circle and straight line then disappear.
- One of the four wavy lines is rotated 90° and now divides the oval. The other three wavy lines disappear.
- One of the top shapes is brought down to the top of the oval and turns from white to black. The spare top shape disappears.
- One of the bottom shapes is brought up to the bottom of the oval and turns from black to white. The spare bottom shape disappears.

1 b

- The figure rotates 90° clockwise each time.

2 a

- The circle moves between the top and bottom of the shape each time.
- The triangle moves downwards and changes colour each time.
- The square moves upwards and changes colour each time.

3 e

- The line with the coloured circle and white circle on the ends rotates 90° clockwise each time.
- The line with the white circle and the smaller line on the ends rotates 90° anti-clockwise each time.

4 b

- 4 of the circles disappear each time.

5 d

- The triangle and the heptagon move 180° around the oval each time.
- The circle changes colour from black, to gold, to white, and then repeats this pattern.
- The outline of the oval changes between a solid line and a wavy line each time.

6 a

- The hexagon and the octagon switch positions each time.
- The octagon switches from its largest size to its middle size, then to its smallest size, and then jumps back to its largest size and repeats the sequence.
- The hexagon switches from its middle size to its largest size, then back to its middle size, then to its smallest size, and then back to its middle size. It never jumps from largest to smallest, or from smallest to largest.

1 c

In all three figures:
- The line intersects both shapes.

2 e

In all three figures:
- There is a green circle and a black circle.
- The main shape has four sides.
- The circles connect to the corners of the main shape.

3 a

In all three figures:
- The shapes are all the same.
- The total number of shapes correspond to the number of sides of this shape.

4 e

In all three figures:
- The square is in the centre of the shapes.
- There is a black circle in the centre of the square.
- The two additional shapes are on opposite sides of the square.

5 b

In all three figures:
- There are three triangles.

6 d

In all three figures:
- The largest shape has seven sides.
- The second-largest shape has no straight sides.
- The smallest shape is a triangle.

1 b

2 d

3 e

4 c

5 a

6 c

1 a

The right column of the grid is a reflection of the left column, using the central vertical line as a mirror line.

The bottom row of the grid is a reflection of the top row, using the central horizontal line as a mirror line.

2 b

In each row, the right column of the grid switches the sizes of the shapes in the left column.

The right column is also a mirror reflection of the left column, using the central vertical line as a mirror line.

The colours of the shapes are switched in each row (e.g., in the first row the largest shape is purple and the smaller shape is white, but in the second row the largest shape is white and the smaller shape is purple).

3 d

The right column is a mirror reflection of the left column, using the central vertical line as a mirror line.

4 c

Looking at each row of the grid from left to right, the white shape moves clockwise around the corners of each box and the black shape moves anticlockwise around the edge of each box.

5 a

Looking at each row of the grid from left to right, each box moves one space right each time.

6 e

Each row is the same as the row above but flipped vertically.

1 b

A = Larger shape is a pentagon. B = Larger shape is a circle. C = Larger shape is a square.

R = Smaller shape is purple. S = Smaller shape is white. T = Smaller shape is black.

2 a

X = Inner shape is a circle. Y = Inner shape is a triangle. Z = Inner shape is a square.

E = Solid lines. F = Dotted lines. G = Diamond patterned lines.

3 c

H = Arrow shape points right. I = Arrow shape points down. J = Arrow shape points left.

Q = Inner shape is a circle. R = Inner shape is a square. S = Inner shape is a rectangle.

4 e

L = Arrow with nothing on its base. M = Arrow with line on its base. N = Arrow with square on its base.

T = All white circles. U = All black circles. V = Black and white circles.

5 d

C = 7 squares total. D = 9 squares total. E = 10 squares total.

R = 2 black squares. S = 1 black square. T = 3 black squares.

6 a

A = Circle is purple. B = Circle is white. C = Circle is black.

X = Arrow is on top. Y = Arrow is on bottom. Z = Arrow is in the middle.